

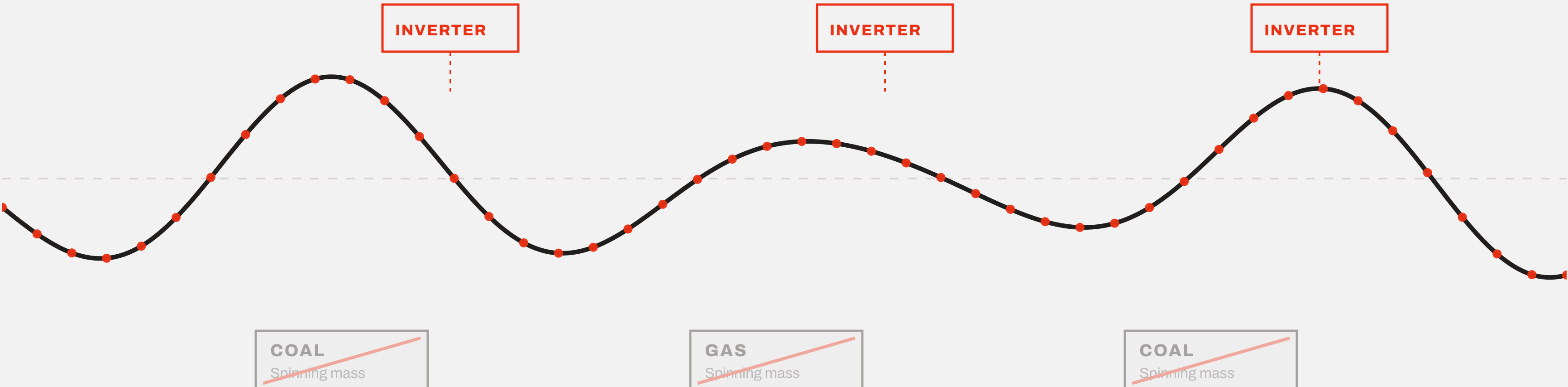


Hybrid PV-Wind System with Grid Synchronisation

Keeping solar and wind in step with a grid that is getting harder to follow



Everything on the grid has to march in step – and the band is losing its drummers



As the heavy machines retire, the grid gets **weaker** – and an inverter’s own output starts moving the very voltage it is trying to follow.

A complete solar-plus-wind plant that stays locked to the grid – and lets go instantly when the grid dies

5 kW of solar and 3 kW of wind, simulated end to end, holding synchronisation while sun and wind swing, and disconnecting within 2 seconds if grid power is lost.

5 kW

Solar array

3 kW

Wind turbine

700 V

Shared DC bus

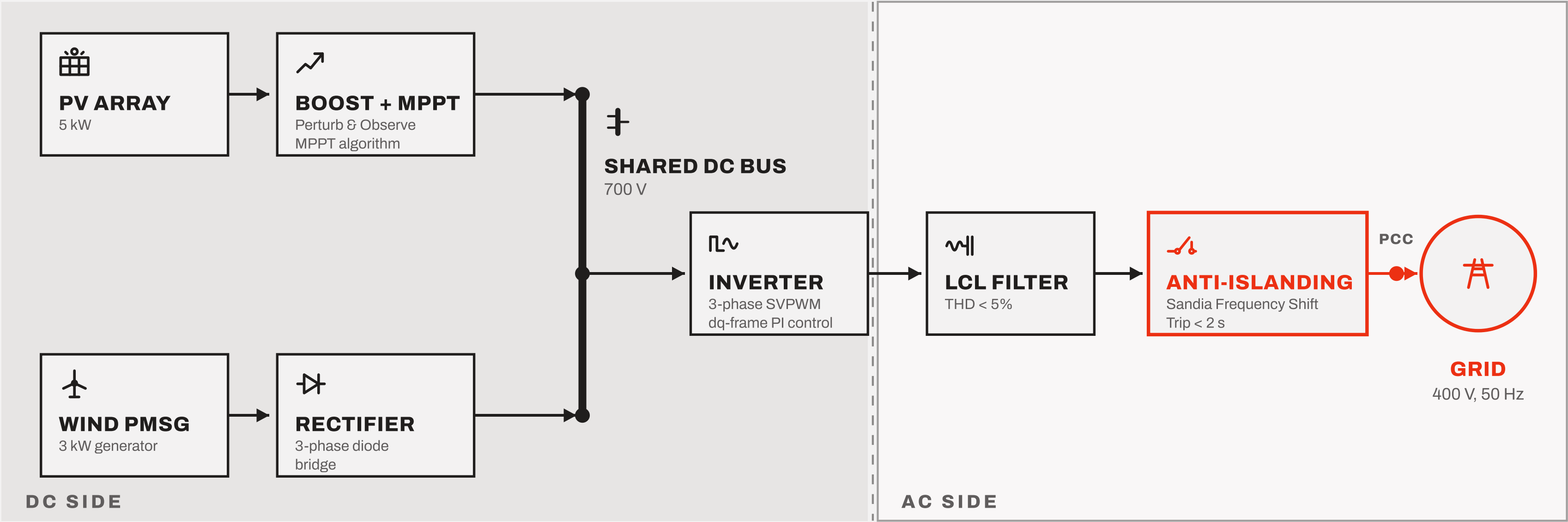
400 V / 50 Hz

Grid connection

100%

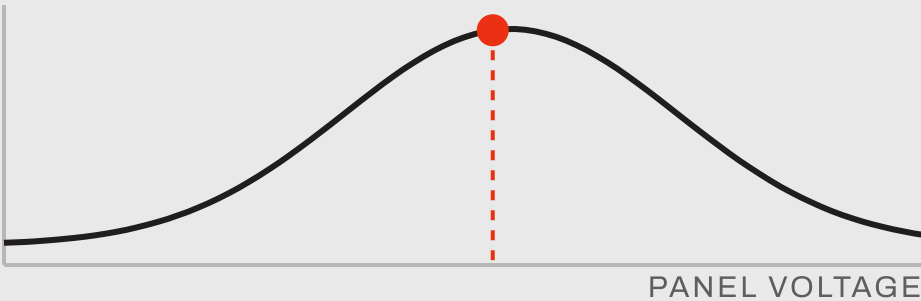
Simulation – MATLAB,
Simulink, Simscape
Electrical. No hardware.

Two sources, one shared DC bus, one connection to the grid



Two sources, one shared 700 V DC bus, one connection to the grid

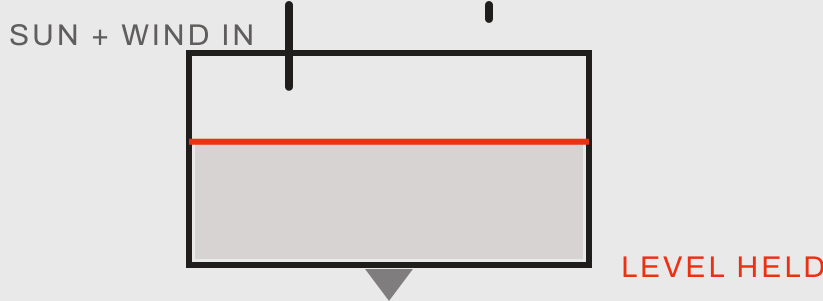
Underneath it all, the controller only has three jobs



01

Harvest.

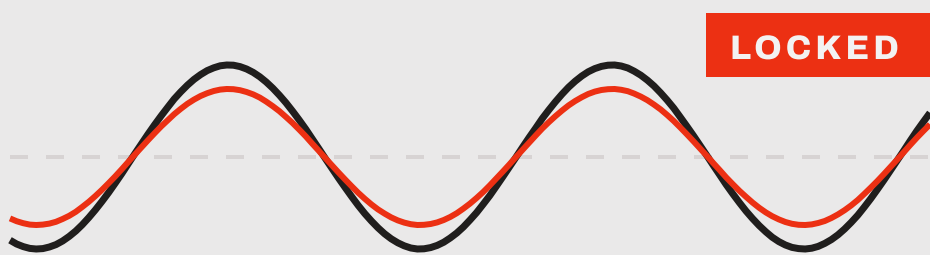
A solar panel has one voltage where it delivers the most power, and that point moves with the sun. Maximum Power Point Tracking (MPPT) hunts for it continuously.



02

Balance.

The 700 V DC bus is a bucket. Sun and wind pour in, the inverter drains it out to the grid. Hold the level steady and the power balances itself – no matter how the sun/wind split changes.



03

Sync.

Before pushing a single amp, we need to know exactly where the grid’s 50 Hz wave is right now. A Phase-Locked Loop (PLL) locks onto that rhythm, and everything else takes its timing from it.

Card 2 is why we chose one shared DC bus. It turns “hybrid” from a hard problem into a bookkeeping problem.

Six numbers decide whether this works

REQUIREMENT	TARGET
Current distortion (Total Harmonic Distortion, THD) at rated output	below 5%
DC bus voltage deviation, and recovery	within 5%, recover in 200 ms
Inner current loop settling	under 2 ms, under 10% overshoot
Phase-Locked Loop re-lock after a 30° phase jump	within 100 ms
Islanding detected and disconnected	within 2 s
Stable operation down to short-circuit ratio (SCR)	SCR of 3

Benchmarked against AS/NZS 4777.2:2020, the Australian standard for inverters connecting to the grid. These are targets, not results.



We are on schedule, and we know where the tight part is

DONE · W4-W6

- architecture locked
- first full-system draft model
- automated test harness running
- first subsystem tested

RUNNING NOW

- grid-sync Phase-Locked Loop
- current-loop tuning maths
- wind model
- anti-islanding framework

NEXT · W8-W11

- DC bus integration
- full system
- weak-grid testing

**Weeks 8 to 11 are a single chain – each step needs the one before it working.
There is one buffer week at W12.**
That is the risk we are managing.

At the halfway point the useful result is not “it works”.

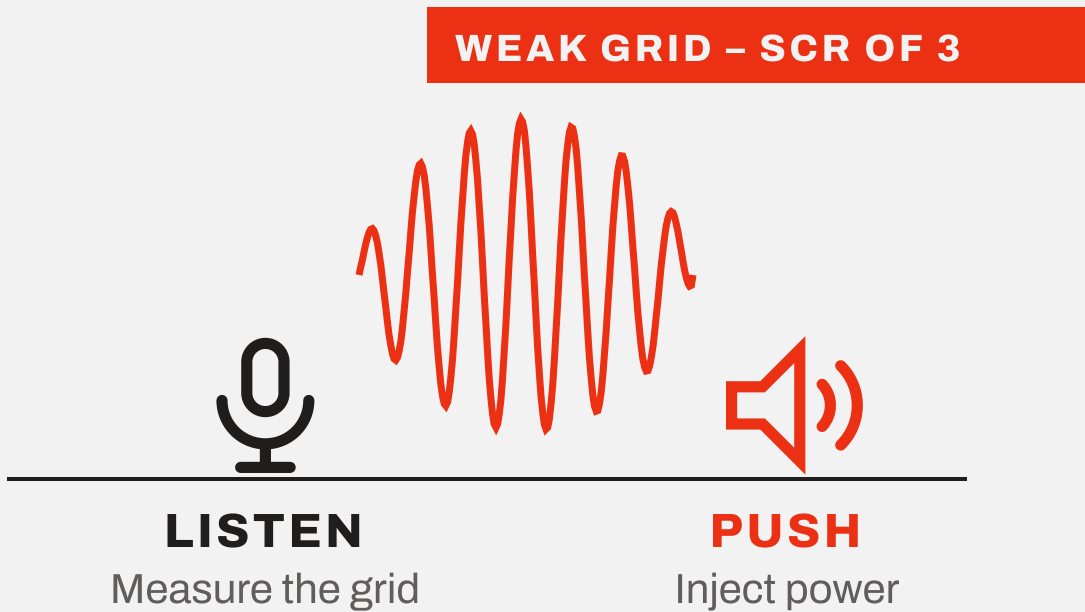
It is a list of things we now know are wrong, while there is still time to fix them.



The model, running

Early-stage Simulink model – live

We are also asking a question most student projects skip – how weak can the grid get before our controller breaks?



It is microphone feedback. Short-Circuit Ratio (SCR) is just the number for how far the microphone is from the speaker.

- 01** A short-circuit ratio (SCR) below about 3 is a weak grid. Our target is to stay stable down to 3.
- 02** It reuses the model we already have – we raise the grid impedance and re-run the same tests. No new subsystem is built.
- 03** It is the test that tells us whether our tuning maths was ever valid, because that maths assumes an infinitely strong grid.

This is our own addition to the scope, not a requirement.

Three things could bite us, and we know what each one costs

- 01

The Week 8–11 chain has one buffer week.

Any slip has almost nowhere to go. Mitigation: subsystems are tested independently first, so integration starts from known-good parts.
- 02

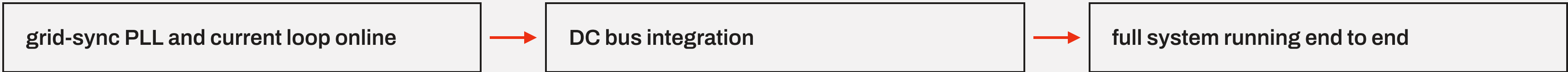
Tuning derived on a simplified model has to survive the real switching model.

Mitigation: the switched model is being built in parallel, not afterwards.
- 03

The solar stage needs a variable-step tracking fix before integration.

Mitigation: already specified – it was finding #3 above.

NEXT THREE WEEKS



Ask us anything – especially if it is not a power question.

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